

Relationship between saturated fatty acids and periodontal disease.



Saturated fatty acids (SFAs) produce an inflammatory response. Hyperinflammation is now recognized as one of the key underlying etiologic factors in periodontal disease. The longitudinal relationship between dietary SFAs and periodontal disease in 264 Japanese individuals, aged 75 years, for whom data were available for the years 2003-2004, was investigated. SFA intake was assessed with a brief self-administered diet history questionnaire. Participants were classified by quartiles of SFA intake. Full-mouth periodontal status, measured as the clinical attachment level (CAL), was recorded at baseline and follow-up examinations. The number of teeth with a loss of $CAL \geq 3$ mm at any site over a year was calculated as 'periodontal disease events'. Poisson regression analysis was conducted, with dietary SFAs as the primary predictor of interest, to estimate their influence on periodontal disease events. High dietary SFA intake was significantly associated with a greater number of periodontal disease events among non-smokers. The multivariate adjusted relative risk (95% confidence intervals) in the 1st, 2nd, 3rd, and 4th quartiles of dietary SFAs was 1.00, 1.19 (0.72-1.97), 1.55 (0.95-2.52), and 1.92 (1.19-3.11), respectively. These findings suggest an independent association of dietary SFA intake to the progression of periodontal disease in older Japanese non-smokers. ABBREVIATIONS: saturated fatty acid (SFA); clinical attachment level (CAL); Toll-like receptor (TLR); lipopolysaccharide (LPS); brief self-administered diet history questionnaire (BDHQ); decayed, missing, and filled teeth (DMFT); clinical attachment level (CAL); body mass index (BMI); relative risk (RR); confidence intervals (CI); nuclear factor-kappa B (NF- κ B).